

EM Simulation 5040

Final Project

Problem 1: 1D Wave Propagation and PEC Reflection

Description: Implement a 1D FDTD simulation of a plane wave propagating in free space, striking a Perfect Electric Conductor (PEC) boundary. Excite the grid using a Gaussian pulse modulated at a center frequency of 10 GHz. Ensure the grid resolution (Δz) satisfies the rule of thumb ($\Delta z \leq \lambda/20$) at the highest frequency of interest.

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The simulation is based on 1D FDTD simulation. Here the 1D has been modified to address the question requirements. The simulation has electric field and magnetic field staggered in space. Electric field is defined at the center while magnetic field is at the edges. E and H fields are updated in the main loop. Following the rule of thumb, Δz was selected with the highest frequency 12GHz. The grid spacing is $\lambda/20$ and the time step is determined by $\Delta t = \Delta z / c_0$ [11].

For boundary, PEC is used in both walls. PEC is implemented at the boundary by not updating electric field at the boundary. i.e $E_z[0] = 0$ and $E_z[-1] = 0$ making both left and right walls PEC [12]. The pulse propagates towards the left and right PEC walls. When the waves hit those walls, the waves are reflected back which we can see on the screenshots with a 180 degree phase shift. This is because, the PEC makes no energy to be absorbed or

transmitted. i.e $E_{total} = E_{incident} + E_{reflected} = 0 \implies E_{reflected} = -E_{incident}$ [10].

For the source, a Gaussian pulse modulated is used. It is a sinusoidal pulse with a central frequency of 10Ghz. Screenshots of the pulse before reflection, during reflection and after reflection is given.

The signal is then converted to frequency domain using FFT(Fast Fourier Transform). By looking at the pulse, the energy is peaked at 10Ghz frequency. The energy is distributed across 8-12Ghz which is the region converging the X band. The Gaussian pulse spreads around the center frequency.

Plot 1: E field snapshot before reflection

At time step 200, the modulated Gaussian pulse is travelling in opposite direction from the source point.

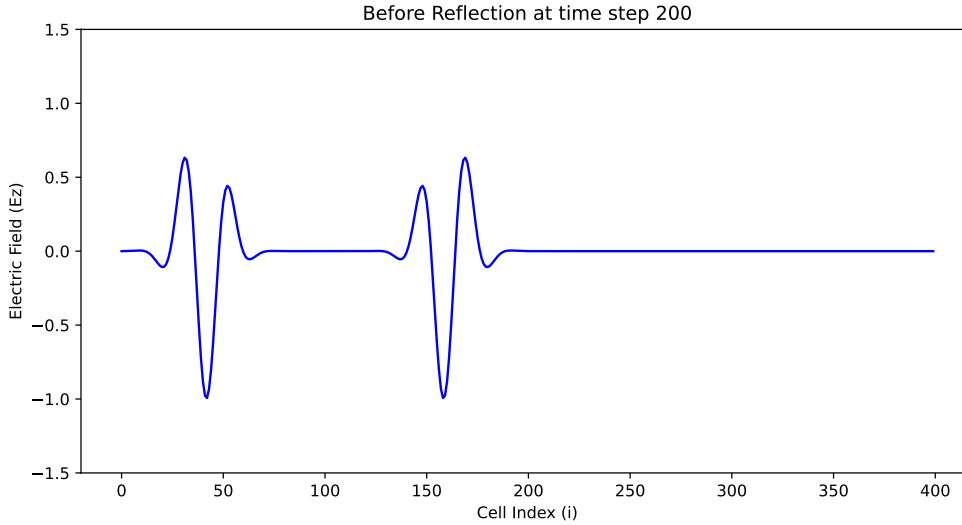


Figure 1: Ez snapshot before reflection

Plot 2: E field snapshot during reflection

The left moving pulse has reached the PEC walls where it experiences 0 electric field. The incident wave (the left pulse) overlaps with its own reflection and we see that pattern.

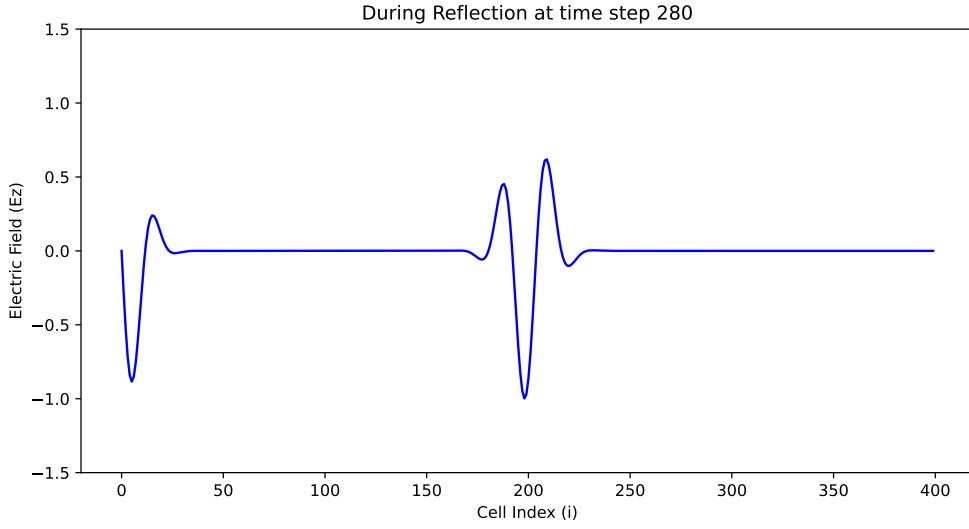


Figure 2: Ez snapshot during reflection

Plot 3: E field snapshot after reflection

Both left and right pulse has fully reflected back after hitting the PEC walls. As mentioned before, the incident and reflected wave will be equal because of which we see that 180 degree phase shift.

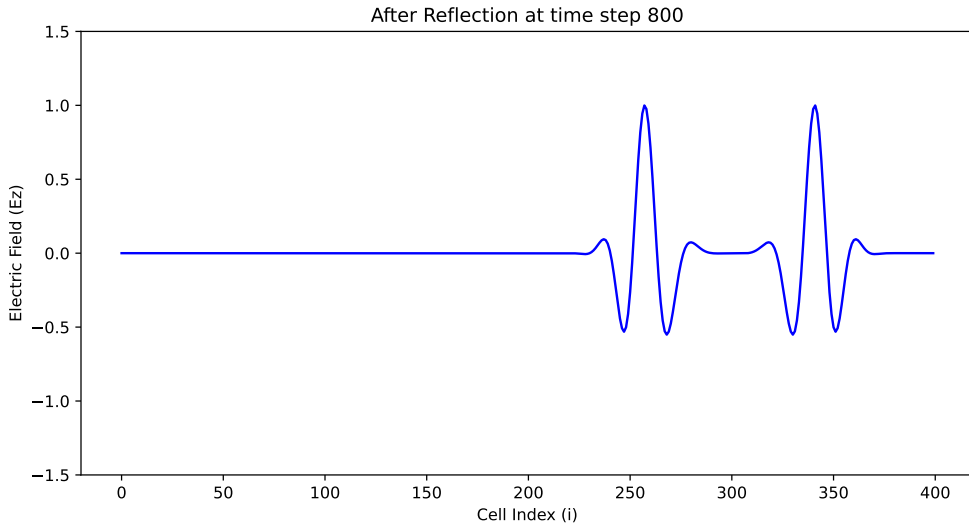


Figure 3: Ez snapshot after reflection

Plot 4: The Frequency spectrum (FFT) of the source pulse, verifying coverage of the X-band

This is the frequency domain plot of the applied source pulse resulting in a smooth bell shape curve. The magnitude is seen maximum at the frequency of 10 GHz. The shaded region represents the X band which also show the center part of energy is inside that region.

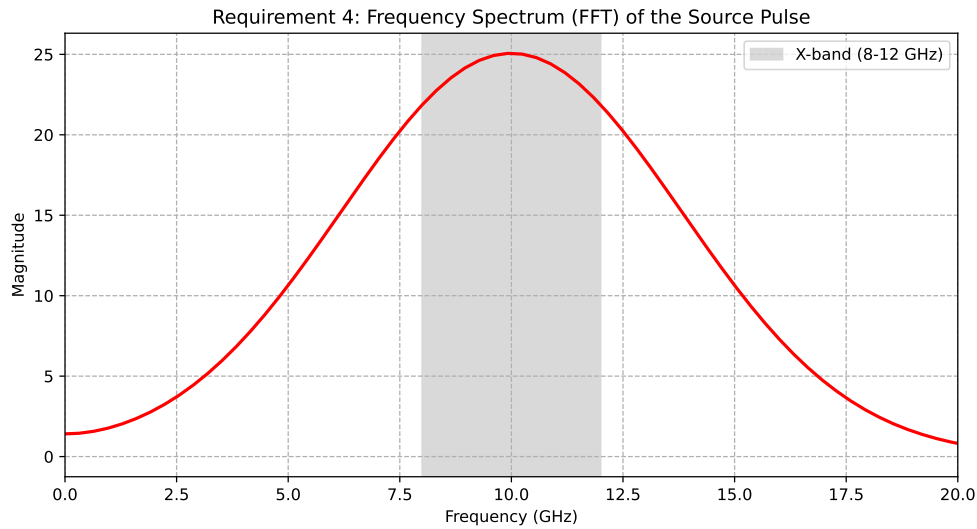


Figure 4: FFT pulse

Problem 2: 2D TE_z Mode Cutoff in a Parallel-Plate Waveguide

Description: Extend to a 2D grid (x, y) . Simulate a parallel-plate PEC waveguide designed with a cutoff frequency of exactly 9 GHz for the first higher-order mode. Excite the waveguide first with a continuous wave (CW) source at 11 GHz, and then with a CW source at 7 GHz. Use simple Mur first-degree Absorbing Boundary Conditions (ABCs) at the open ends.

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The simulation is using 2D grid waveguide. The fdtd uses Absorbing boundary conditions at the open ends and use the PEC in the other parts. For the source, continuous wave is used: first with 11 GHz frequency and then at 7 GHz frequency. The f_{source} in the code can be changed to switch between 7 GHz or 11 GHz frequency. The fdtd domain is divided into small cells and the electric and magnetic field are updated alternatively in time [12]. The same E and H field update is used as in other problem simulations as well. Here the grid spacing is not defined based on free space wavelength but by the waveguide geometry i.e physical width for dominant mode. i.e $f_c = c_0/2a$ [10] Additionally, the grid space has been reduced here as well giving a finer mesh and high spatial resolution. 2D snapshots of the E field magnitude for the 11 GHz and same for the 7 GHz has been provided. Also a measurement point has been setup far away from the source to capture the time domain signal. The measurement has been done for both 7 GHz and 11 GHz.

We need to change the frequency from 7 GHz to 11 GHz in the $f_{source} = 11.0 \times 10^9$.

As we know [10] if frequency $\geq f_c$, then the wave is travelling wave i.e wave propagates. Similarly, if $f < f_c$, then the wave is evanescent wave i.e the wave becomes a decaying standard wave. If we look at the probe measurement and snapshot, we can see the field decaying.

Plot 1: Snapshot of 11 GHz

It is the propagating mode because 11 GHz is higher than the 9 GHz cutoff frequency. We can see the black color in top and bottom walls ($y = 0$ and 60) due to PEC being implemented there. We can see a regular pattern of the wave which might be representing propagation.

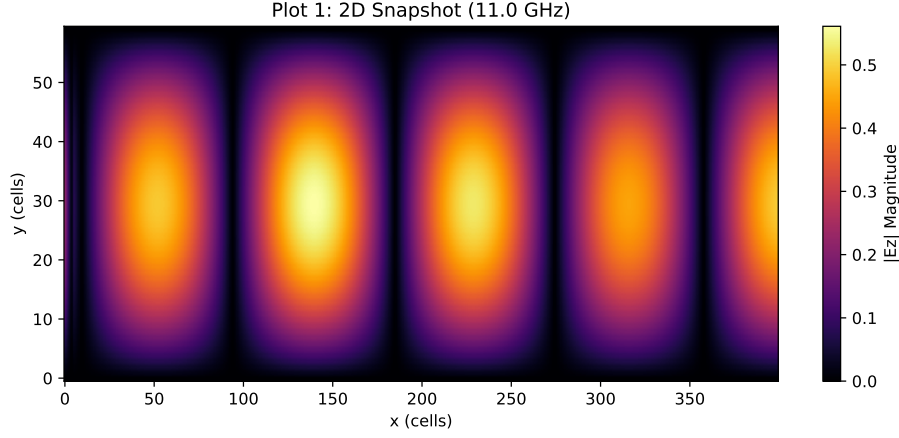


Figure 5: Snapshot of 11 GHz

Plot 2: Probe measurement 11 GHz

The signal starts at 0. After reaching around 300 time steps, the wave builds up. Looking at the pulse pattern, we can see a sine waveform propagating.

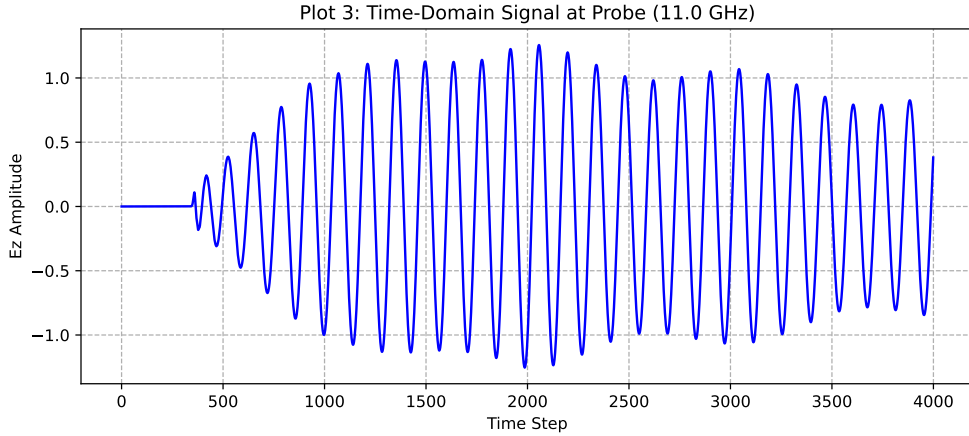


Figure 6: Probe measurement 11 GHz

Plot 3: Snapshot of 7 GHz

We can see at the starting we can see the wave trying to propagate. But the magnitude drops which is similar to what an evanescent wave would

behave. As you move more along the propagation direction, the magnitude gradually fades away.



Figure 7: Snapshot of 7 GHz

Plot 4: Probe measurement 7 GHz

The plot starts with 0 magnitude at the starting point similar to 11 GHz. So as the wave propagates, the wave almost drops to 0 after 1000 time steps.

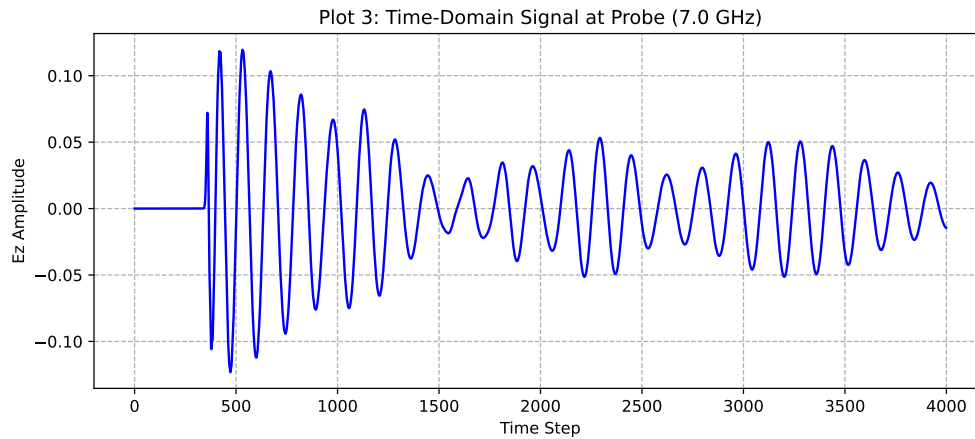


Figure 8: Probe measurement 7 GHz

Plot 5: Phase velocity plot

It shows the analytical phase velocity vs the simulated one. The red indicates the simulated one and the black curve is the analytical one.

For the phase velocity,

$$v_p = \frac{c_0}{\sqrt{1 - \left(\frac{f_{cutoff}}{f}\right)^2}}$$

For analytical we use $f = f_{vec}$.

The theory and simulated phase velocity seems to be offset by some margin.

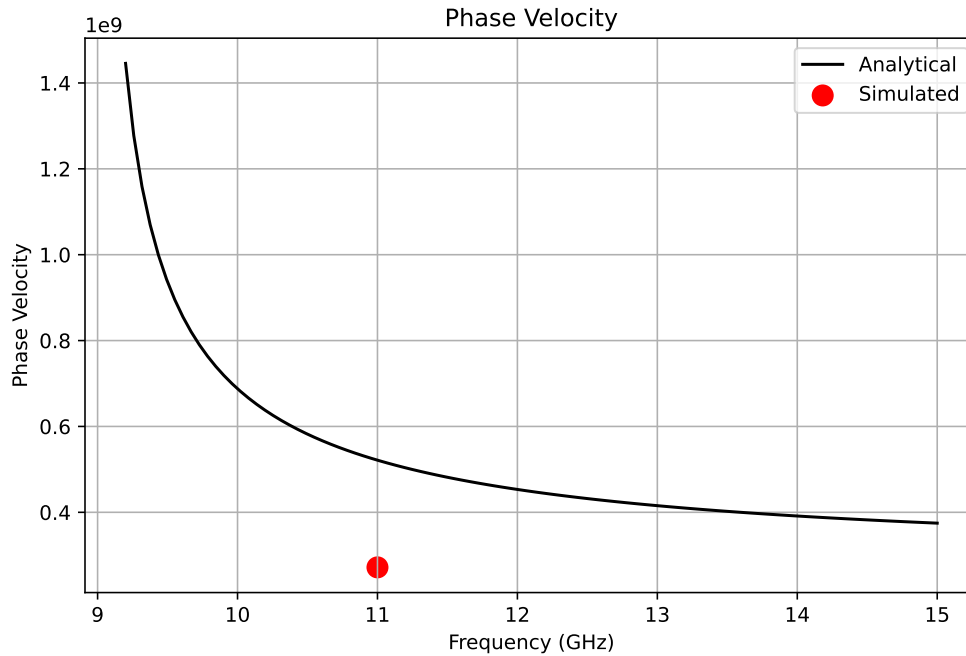


Figure 9: Phase velocity

Problem 3: X-Band Waveguide Iris and S-Parameters

Description: Using the 2D TE_z code, insert a PEC iris (a partial blockage) halfway down a rectangular waveguide. Excite the waveguide with a broadband differentiated Gaussian pulse. Record the incident, reflected, and transmitted time-domain fields using appropriate probe locations.

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The same 2D TE_z has been used as the baseline for this part. So same PEC and ABC [12] has been used as before and other variables has been kept similar as well. Two probes are used for the measurement. The probe for the reflected wave is placed at $x = 80$ which is located between the source and the iris. The probe for the transmitted wave is at $x = 300$ which is behind the iris. Iris is placed at the center $Nx/2$. The iris is a metallic discontinuity that causes partial blockage placed halfway down the rectangular waveguide [13]. Similarly like the Problem 2, here the grid spacing is not defined based on free space wavelength but by the waveguide geometry i.e physical width for dominant mode. i.e $f_c = c_0/2a$ [10] Also the grid space has been decreased giving a finer mesh and high spatial resolution. For the source, the simulation uses a differentiated Gaussian pulse. The S11 and S21 are using dB for the magnitude plots which represents how much power is transmitted or reflected.

Plot 1: 2D E_z field snapshot showing the wave diffracting through the iris

The plot shows the diffraction of the wave through the iris. The top and bottom is seen black due to same PEC used before. The field is seen strongest at the center of the wave. As we can see at 250 we can see discontinuity which is where the iris is placed i.e at the center $Nx/2$. That iris causes the wave to squeeze causing the diffraction in there which is visible to the right side of the plot. The wave once passes that gap, acts as a new point source (Huygens principle).

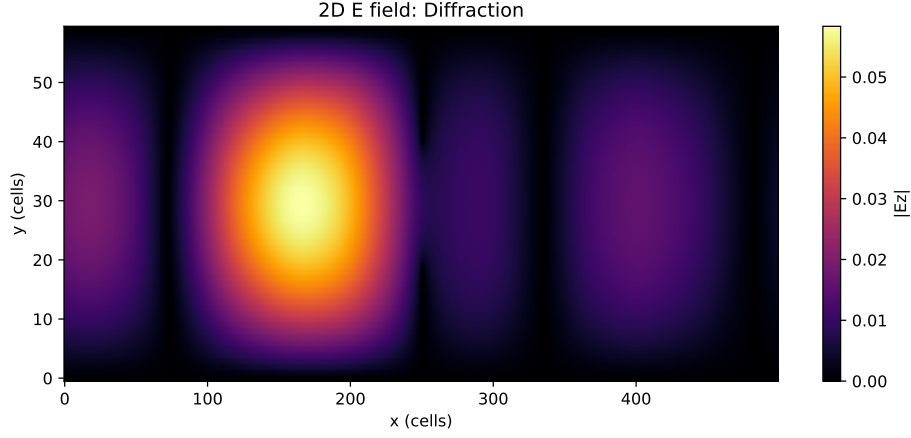


Figure 10: 2D E_z field snapshot: Iris Diffraction

Plot 2: Time-domain signals at the reflection and transmission probe points

The blue curve is the reflection curve which shows early response as its placed at $x = 80$. We see a large oscillation at the first, and then some fluctuations are seen. The overlapping at time steps between 500 and 1000 might be due to incident and reflected wave. The orange curve represents the transmission curve. As we can see a straight line until it hits the time step of around 500. This means the wave has not reached the transmission probe yet. The reason why orange plot is seen smaller than blue one might be that the iris is blocking most of the wave and only small part is transmitting.

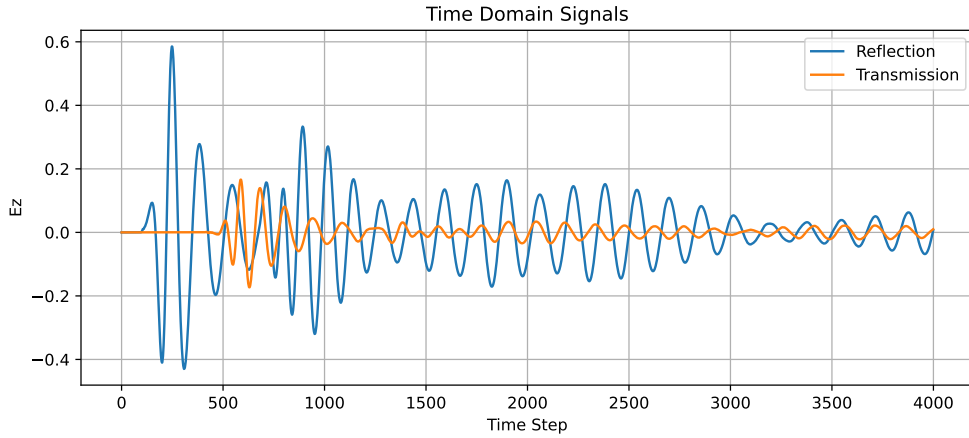


Figure 11: Time-domain signals

Plot 3: S11 return loss

This represents reflected power i.e how much energy bounced. S11 is relatively high which means most of the energy came back reflected.

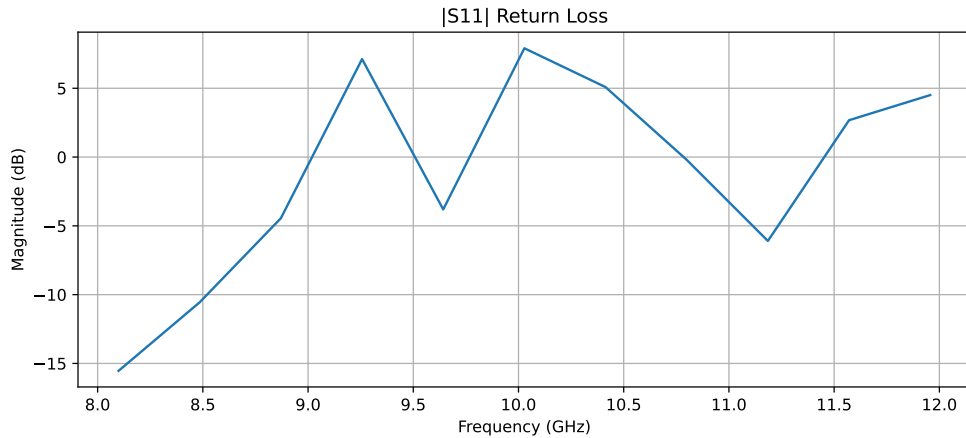


Figure 12: S11 return loss

Plot 4: S21 insertion loss

It represents the transmitted power i.e how much energy reached the other side of iris. We can see that very minimal energy was transmitted. -10 dB transmitted means only 10% of power was transmitted.

$$\text{dB} = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

$$\frac{P_{out}}{P_{in}} = 10^{-10/10} = 0.1 = 10\%$$

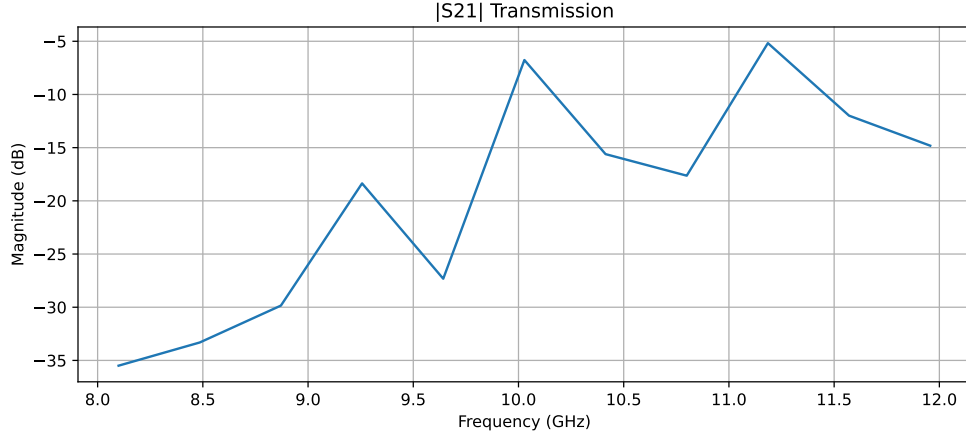


Figure 13: S21 insertion loss

Problem 4: 1D Fresnel Reflection at an SOI Interface

Description: Set up a 1D FDTD domain containing a vacuum-to-silicon ($n \approx 3.48$) interface. Excite the vacuum region with a broadband pulse centered around 192 THz ($\lambda_0 \approx 1.55 \mu\text{m}$). Calculate the numerical reflection and transmission coefficients and compare them against analytical Fresnel equations.

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The simulation consists of a 1D FDTD with a vacuum to silicon interface. The frequency is 192 THz with a corresponding wavelength of around $1.55 \mu\text{m}$.

The numerical and analytical reflection and transmission coefficients are calculated as per the question.

Here in the simulation, there has been use of 2 different loops. The first loop consist of vacuum only and we record the pulse there which will give us incident field. Later, in the second loop, we introduce the silicon interface and we do the same i.e measurement is done which will give total field. So by subtracting the total field with incident field, we get the reflected wave. A transmission probe is also used to record the transmission pulse. The source used here is a broadband pulse centered around 192 THz. A mask filter is used to filter the frequency ($\text{freqs} \geq 180 \times 10^{12}$) & ($\text{freqs} \leq 200 \times 10^{12}$).

During the calculation of numerical reflection and transmission, we are using ratio of silicon and air refractive index for transmission as,

$$R_{\text{numerical}} = \left(\frac{|\text{FFT}_{\text{refl}}|}{|\text{FFT}_{\text{inc}}|} \right)^2$$

$$T_{\text{numerical}} = \left(\frac{n_{si}}{n_{air}} \right) \left(\frac{|\text{FFT}_{\text{trans}}|}{|\text{FFT}_{\text{inc}}|} \right)^2$$

For reflection, no refractive index factor is needed because both waves are in the same air medium. But for transmission, it must use ratio of n_{si}/n_{air} in order to ensure energy conservation as $R + T = 1$ [10].

And for analytical part, we are using Fresnel equations as [10],

$$R_{\text{analytical}} = \left(\frac{n_{air} - n_{si}}{n_{air} + n_{si}} \right)^2 = 0.306$$

and [10],

$$T_{\text{analytical}} = 1 - R_{\text{analytical}} = 0.694$$

Plot 1: Dielectric profile (ϵ_r vs grid index) overlayed with a snapshot of the transmitted E -field

The dielectric interface is set at $nx/2$, which is $x = 600$ which we can see in the plot. Before that interface, the medium is air. Past that interface we can see the blue line shifting around value of around 12 which confirms it is silicon. The E_z is scaled by +5, so that we can see the pulse shape.

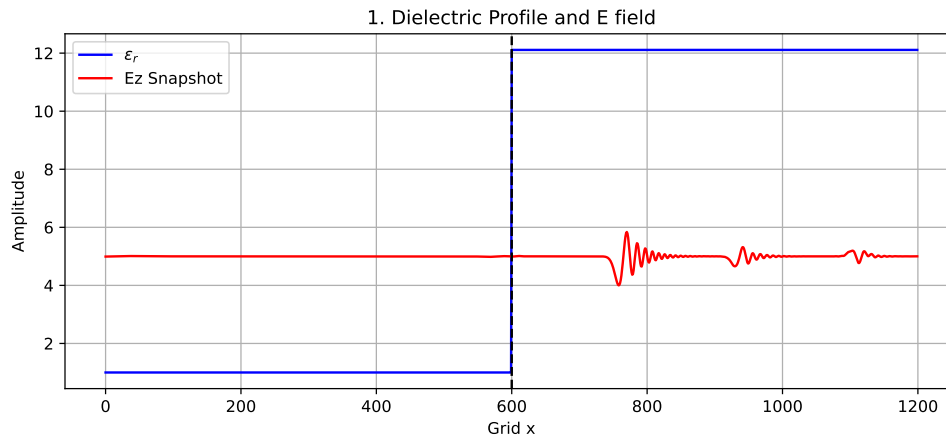


Figure 14: Dielectric profile and E field

Plot 2: Time-domain signal of the reflected pulse

This contains the pulse of the reflected wave that we got from the subtraction. By looking at the pulse, we can see that the pulse is decaying over time. This is the result of ABC at the boundary. Without ABC, we would not see the wave being reflected and there will be no decay. I have attached the snapshot of that as well.

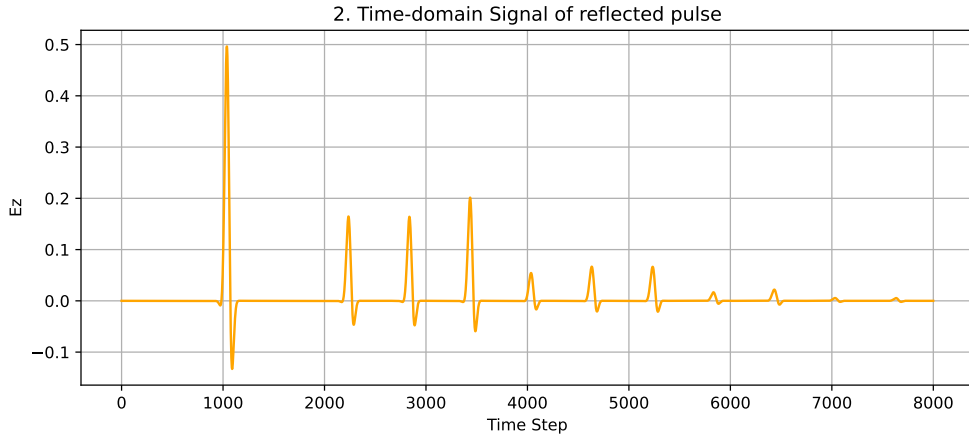


Figure 15: Time-domain signal of the reflected pulse

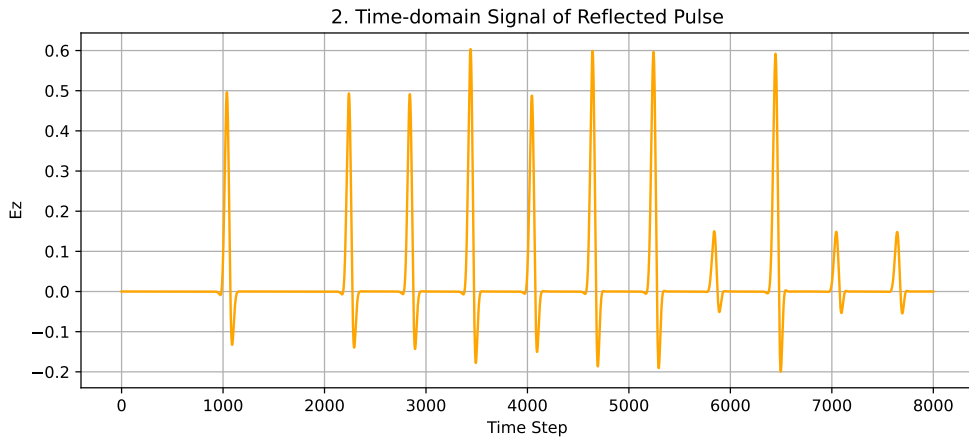


Figure 16: Time-domain signal of the reflected pulse (without ABC)

Plot 3: Computed Reflection Coefficient (R) vs. frequency (180–200 THz)

The numerical and analytical reflection coefficient has been plotted here. The calculation are given above with a value of 0.306 which is why we are seeing a flat line with that value. The analytical and the numerical is seen close at around 192 THz frequency.

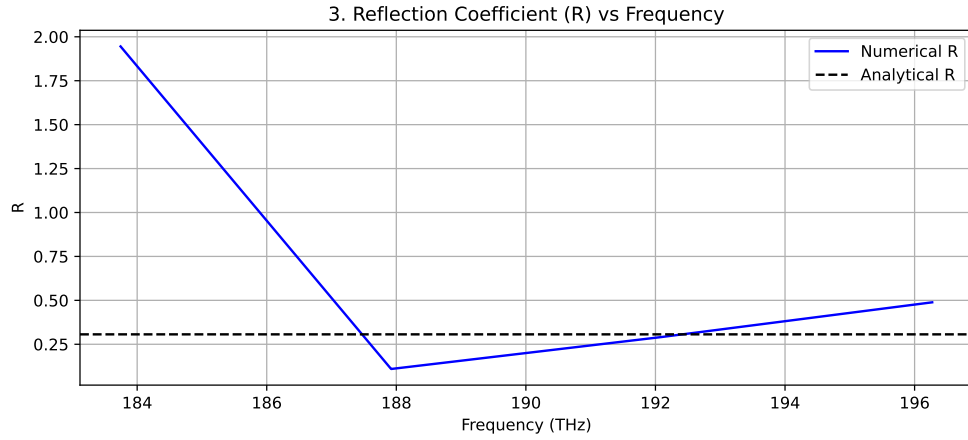


Figure 17: Reflection Coefficient vs Frequency

Plot 4: Computed Transmission Coefficient (T) vs. frequency (180–200 THz)

Similarly, the numerical and analytical transmission coefficient has been plotted here. If we calculate the value of transmission coefficient, we get around 0.694. That means around 69% of the energy was transmitted.

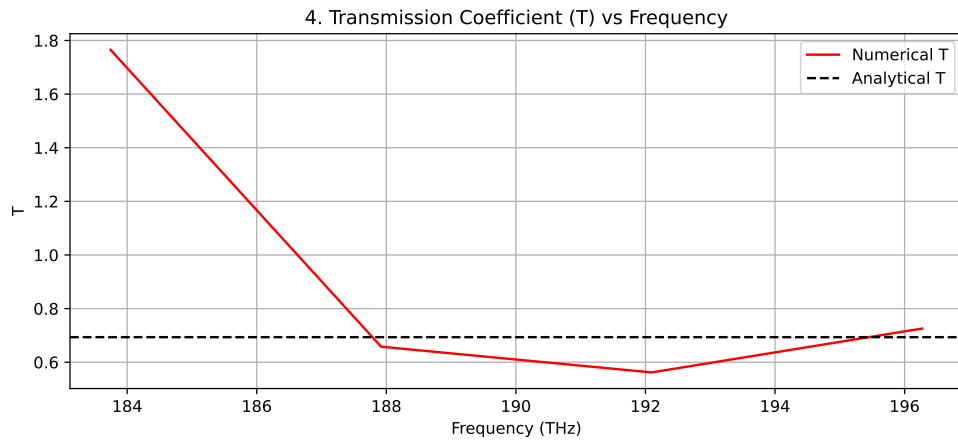


Figure 18: Transmission Coefficient vs Frequency

Problem 5: 2D SOI Slab Waveguide Mode Propagation

Description: Simulate a 2D SOI slab waveguide core (Silicon, $n = 3.48$) surrounded by a cladding (Silicon Dioxide, $n = 1.44$). Calculate the fundamental TE transverse mode profile analytically, and use it as a "hard source" spatial envelope at the input boundary, oscillating at 192 THz.

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For this simulation we are using Silicon-on-Insulator (SOI) slab waveguide. The silicon core with refractive index $n_{\text{core}} = 3.48$ is surrounded by cladding of refractive index $n_{\text{clad}} = 1.44$ as required by the question.

For the source, a fundamental TE transverse mode profile has been used injected as a hard source. With this source, we get a sinusoidally oscillating field in the core and exponentially decaying field in the cladding.

The equation for the modal profile is [2]:

$$E = \begin{cases} A \cos(\kappa y) & \text{(inside the core)} \\ A \cos(\kappa d) \exp(-\gamma(|y| - d)) & \text{(at the cladding)} \end{cases}$$

Here κ is the transverse wavenumber in the core, γ is the decay constant in the cladding, and β is the phase constant [2].

$$\kappa^2 = n_1^2 k_0^2 - \beta^2$$

$$\gamma^2 = \beta^2 - n_2^2 k_0^2$$

where the free-space wavenumber $k_0 = 2\pi/\lambda_0$, and the refractive index of the core and the cladding are respectively n_1 and n_2 .

For β , we have [2]:

$$n_{\text{eff}} = \beta/k_0$$

So for the calculation of n_{eff} (effective index) to be used for the calculation

of β in the equation, we use the transcendental characteristic equation [2]:

$$V\sqrt{1 - b_{\text{TE}}} = 2 \tan^{-1} \left(\sqrt{\frac{b_{\text{TE}}}{1 - b_{\text{TE}}}} \right)$$

The n_{eff} calculation from EIM is taken from paper [2] and the code is inspired from [9]. We can refer to [2] equation 97, 98, 99 for full equations.

The waveguide has a transverse grid of $N_y = 120$ out of which there is a 10 cell silicon core with remaining as cladding silicon dioxide. The simulation fills the whole waveguide with cladding first and then inserts the core region.

For the Poynting vector, we are using the equation:

$$S = -0.5E_zH_y$$

The cross section is taken by using vertical slice of the 2D grid at the midpoint ($N_x/2$) in the code.

Plot 1: The analytical transverse E -field amplitude profile applied at the source plane

It represents the fundamental TE mode. The plots explains that the wave is cosine shape in the core (this is where the light is confined). The outside region is the cladding where the curve decays exponentially.

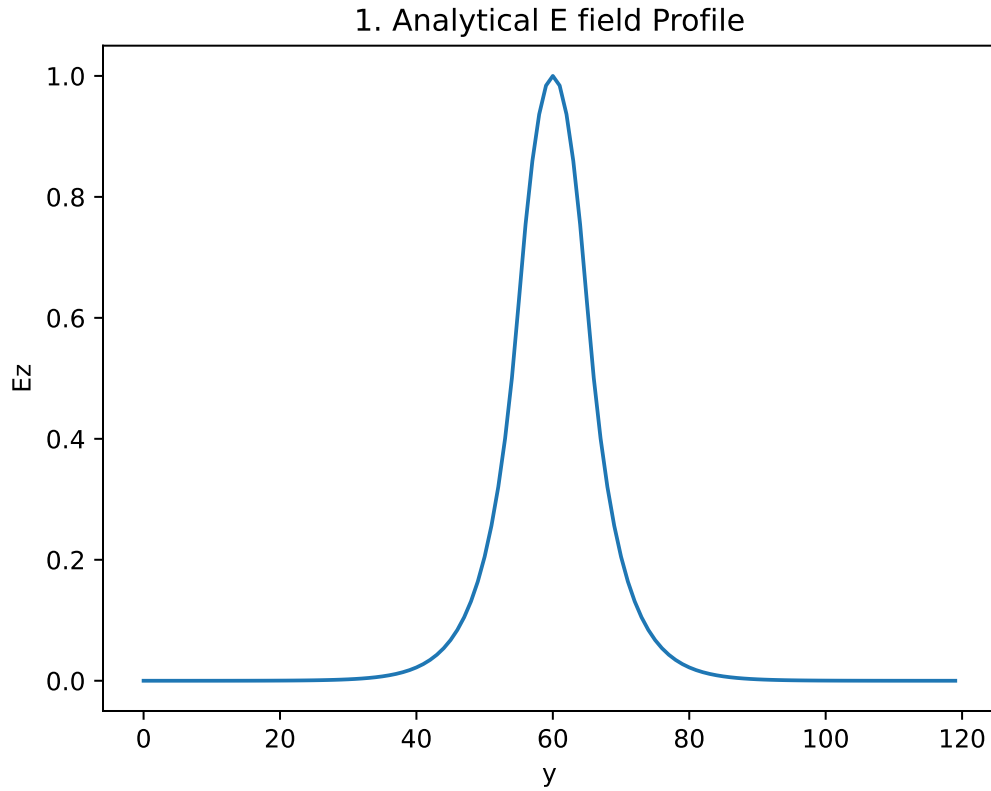


Figure 19: Analytical transverse E -field

Plot 2: 2D snapshot of the E -field

We can see that the red and blue (the positive and negative peak) of the waves are lined at $y = 60$. This means that the wave has been confined into core and not leaked into cladding. We can see a uniform and continuous wave along x , which might be an indication of the steady state.

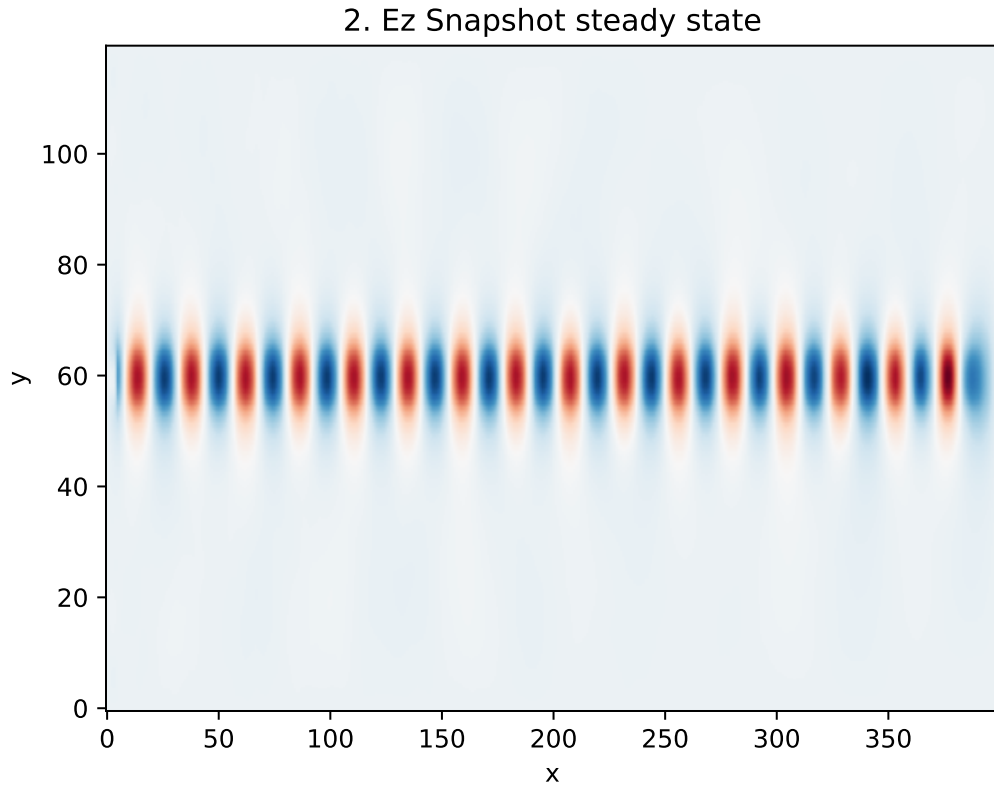


Figure 20: 2D snapshot of the E -field

Plot 3: 2D snapshot of the phase front

This is a zoomed in picture which makes the plot to analyze easily. As we can see the red and blue bars are very close to each other meaning there is less gap between the alternating peaks. This happens due to wavelength compression. The fade out in the top and bottom represents the cladding exponential decay.

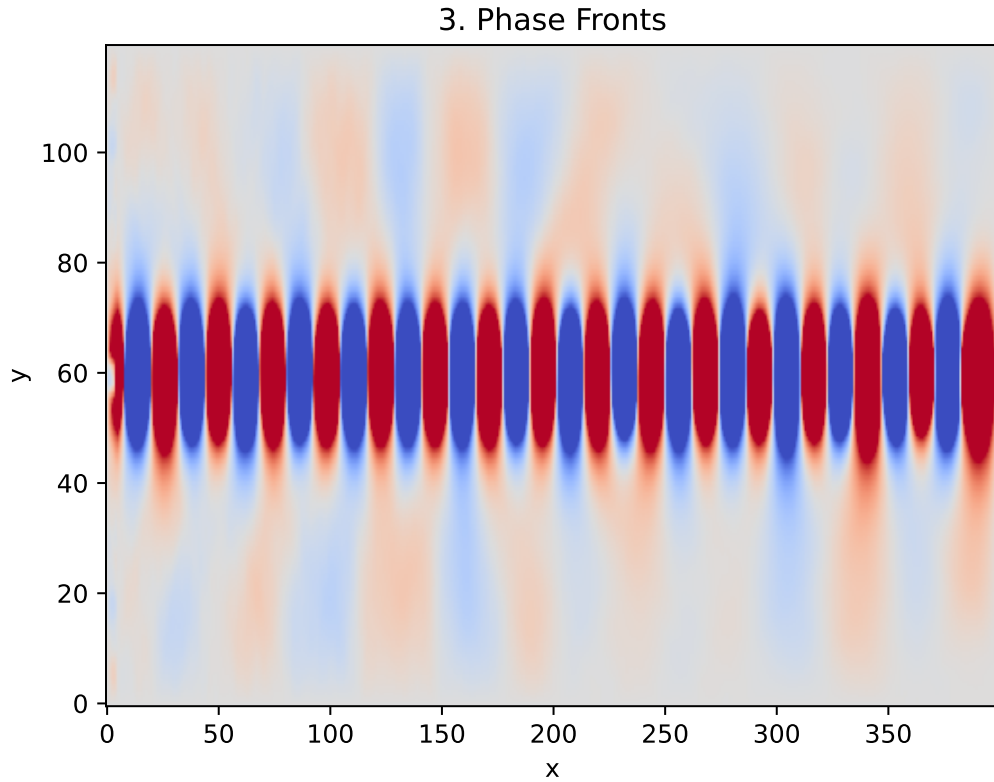


Figure 21: 2D snapshot of the phase front

Plot 4: Poynting vector

The Poynting vector is calculated as $S = E * H$. S_x means the energy travelling towards x axis. The electric field modal profile and the power curve looks kind of similar but in the power plot, we can actually see the energy more concentrated in the middle giving that sharp look comparatively.

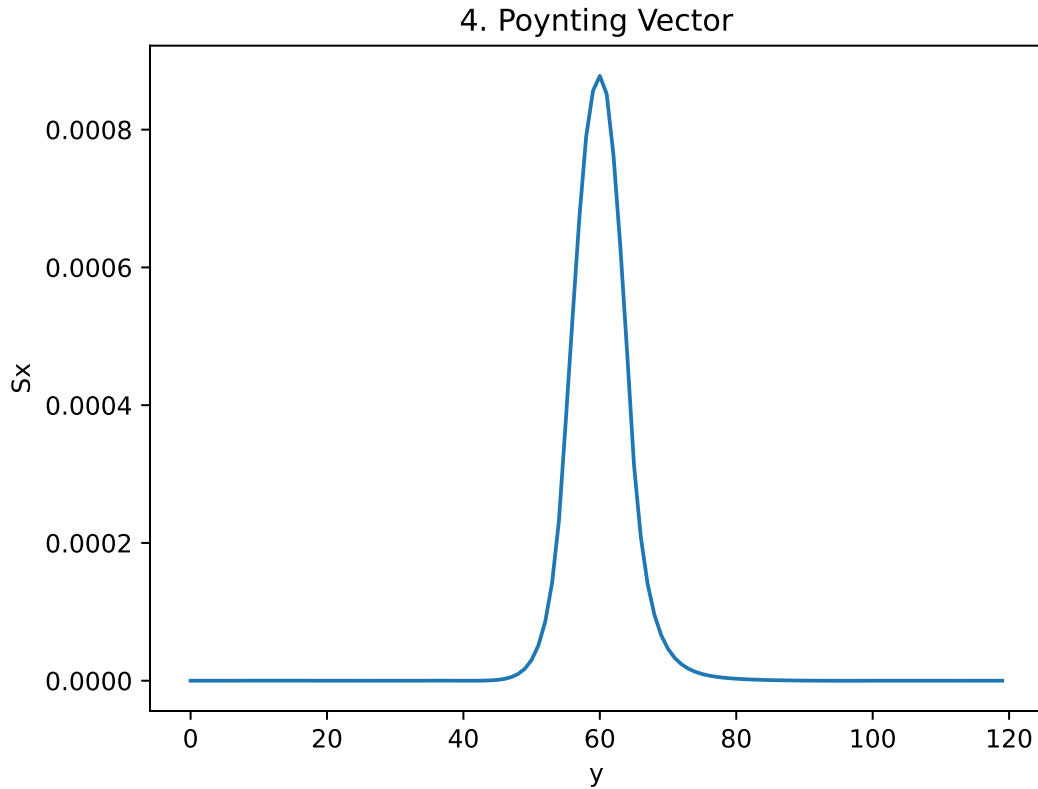


Figure 22: Poynting vector

Problem 6: SOI Waveguide Micro-Bend Radiation

Description: Modify the previous 2D waveguide to include a 90-degree bend. The bend radius should be small enough to induce visible radiation loss into the cladding.

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The FDTD modifies the previous 2D waveguide to simulate a SOI waveguide to include a 90 degree bend. So it captures how the waveguide interacts with the 90 degree discontinuity [8].

For the bend part, the light travels horizontally along $y = 100$ and then the light hits the 90 degree bend at $x = 250$. After that, the light travels vertically along $x = 250$. So the coordinate for the bend is located at $(x = 250, y = 100)$.

For the measurement of the time domain signal, a probe is placed after the bend which corresponds to coordinate of $(x = 250, y = 450)$.

ABC boundary is also utilized in the walls to absorb reflections. We can see the difference of using ABC and not using in plot 1 snapshots.

The time domain signals are first converted into frequency domain and by taking a ratio of $E(f)_{\text{out}}/E(f)_{\text{in}}$ and squaring their magnitude we get efficiency.

Energy density is the total energy stored in the E and H fields. The equation for energy density is:

$$\text{energy_density} = 0.5 (\epsilon_0 \epsilon_r E_z^2 + \mu_0 (H_x^2 + H_y^2))$$

Plot 1: 2D snapshot of the E -field

The plots illustrated the 2D snapshot of the E field through the bend of the L shape. The outside ripples that we see is the resulting radiation change. Without ABC we see lots of wave reflections that creates undefined patterns.

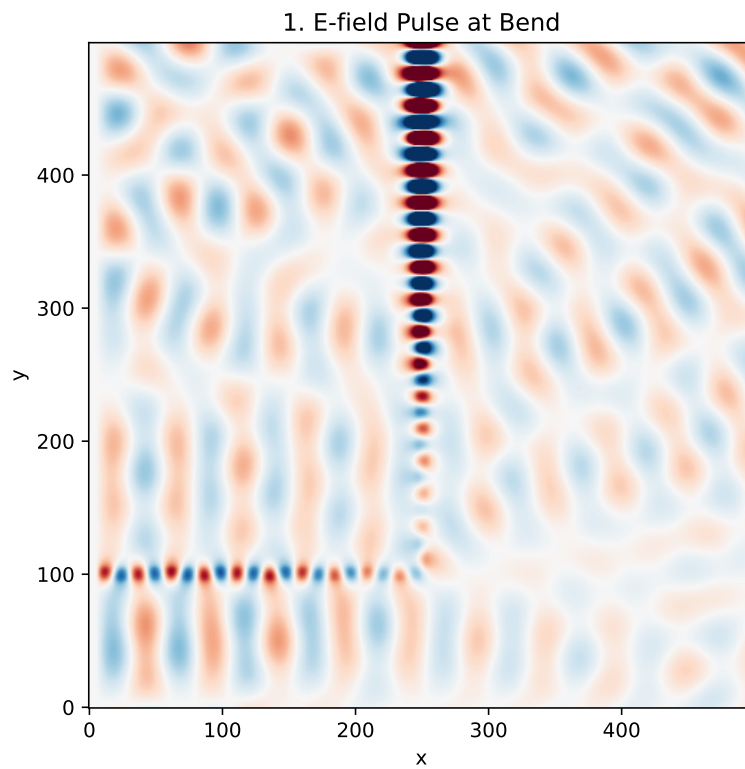


Figure 23: 2D snapshot of the E -field

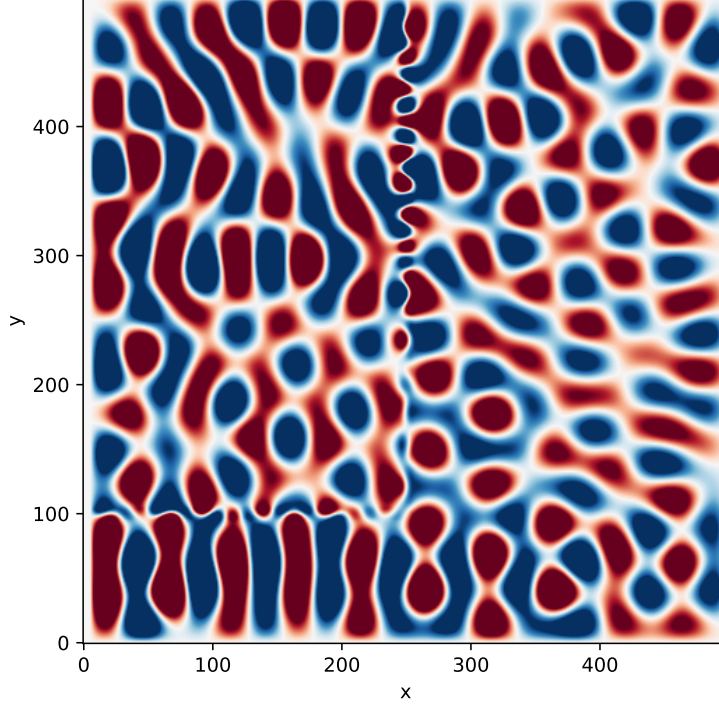


Figure 24: 2D snapshot of the E -field(Without ABC)

Plot 2: Time-domain signal at a transmission probe located after the bend

The plot displays the time domain signal at a probe located after the bend. Initially we see a flat line until around 170 fs time and then we see oscillations begin. This is due to the light has to travel horizontally and then vertically through the bend to reach that probe point where we are measuring. The peak of the signal is seen around 0.8.

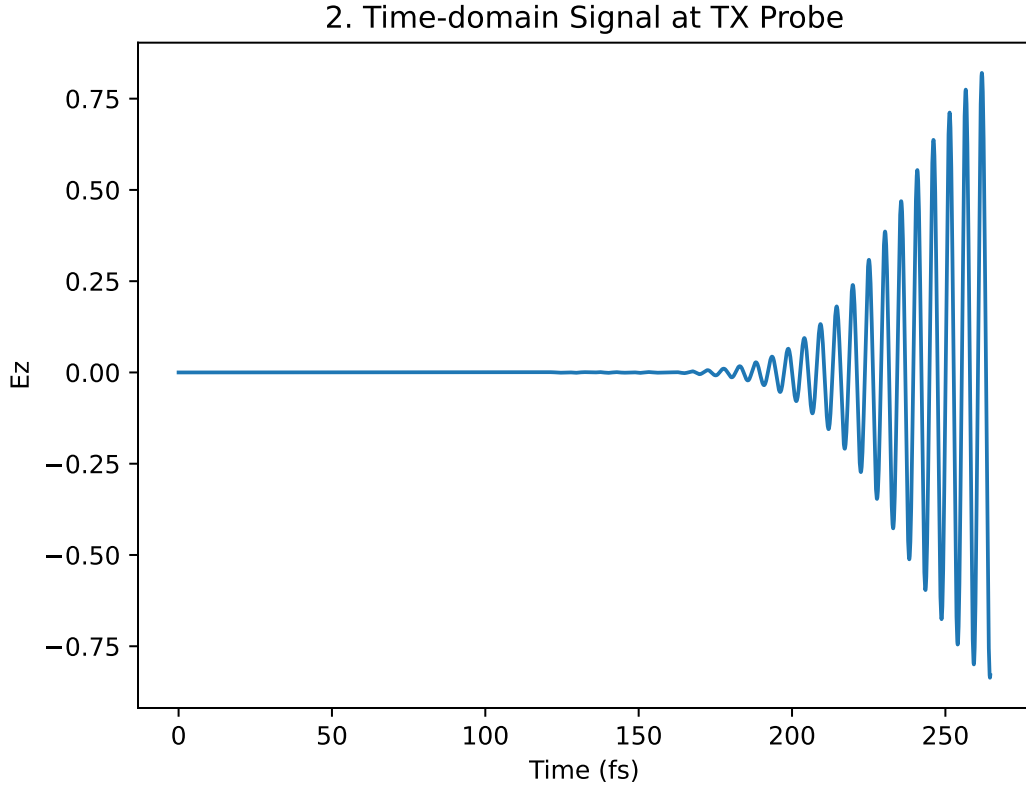


Figure 25: Time-domain signal at a TX probe after the bend

Plot 3: Transmission efficiency (Output Power / Input Power)

The plot shows transmission efficiency between (180–200 THz) frequency. At around 181 THz, we see about 70% of energy whereas at 189 THz, only around 10% of energy can be seen followed by 30% at around 200 THz.

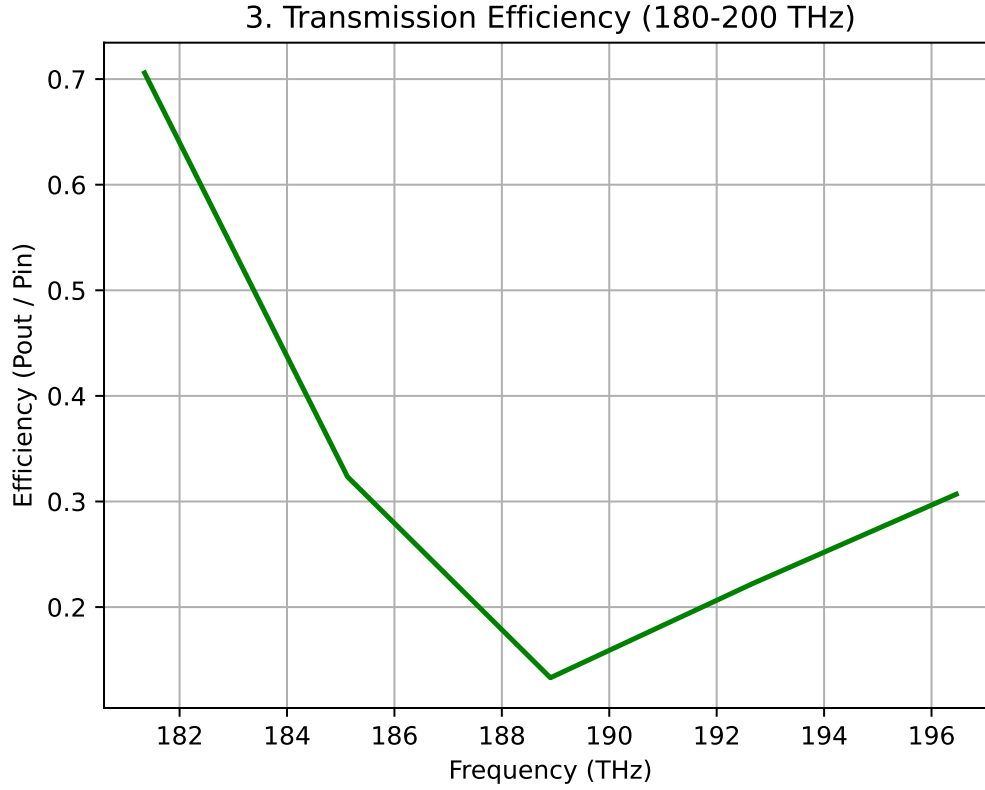


Figure 26: TX Efficiency Pout/Pin

Plot 4: Energy density

Here a log scale is used for the energy density map. Earlier linear scale was used, but there was no visible leakage seen. So I found out that using log scale and adding 1×10^{-15} for reducing error helps to visualize the leakage radiation. As we can see, the orange glow outside the bend is the radiation leakage pattern. The reason for that maybe is the breaking of total internal reflection after that bend.

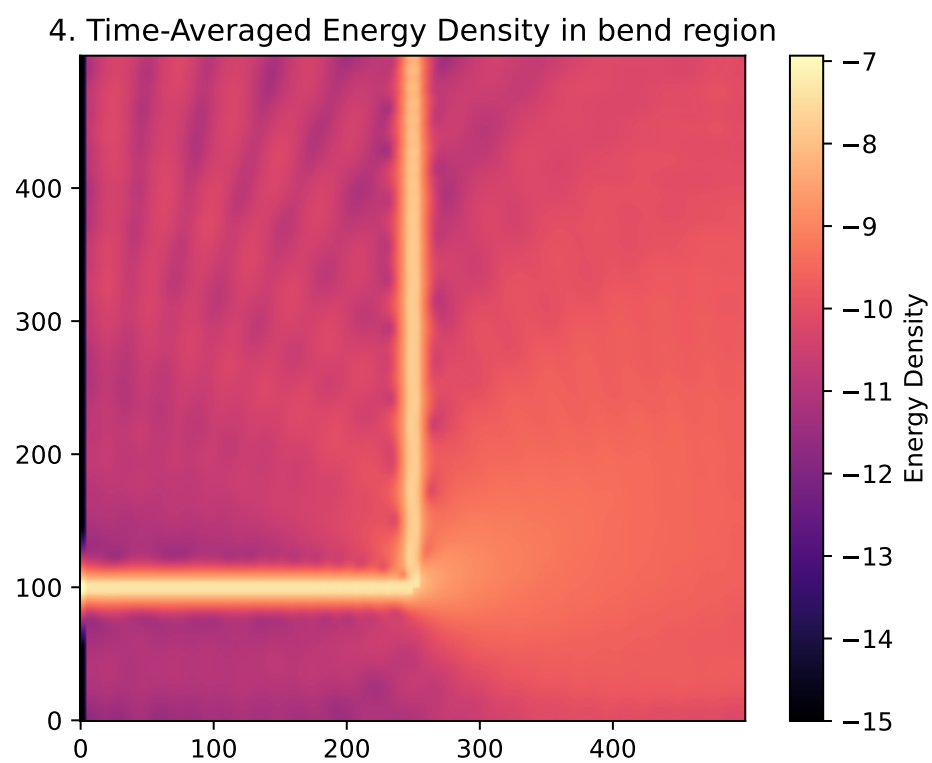


Figure 27: Energy density

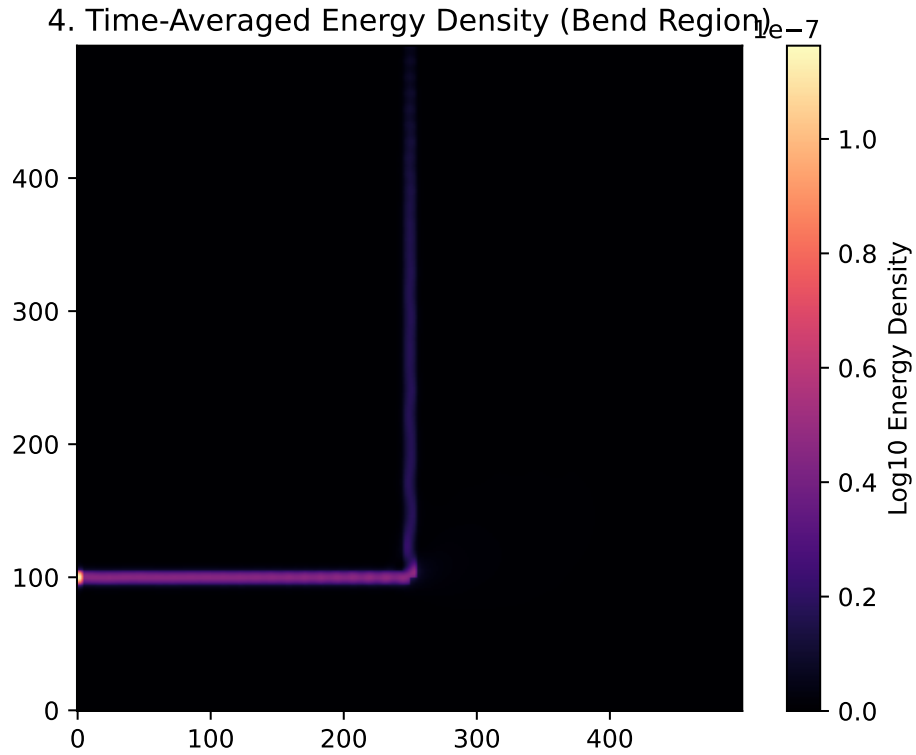


Figure 28: Energy density(Linear Scale)

Problem 7: 2D Radiation from a PEC Slit Aperture

Description: Create a 2D domain separated into two halves by a PEC wall with a central slit. Excite the slit with a local current source or a localized waveguide feed. Implement a basic Uniaxial Perfectly Matched Layer (UPML) or Convolutional PML (CPML) on the outer boundaries to simulate free-space radiation without artificial reflections.

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This FDTD implements a very basic CPML (Convolutional Perfectly Matched

Layers) and also introduces a slit separated by two parts of the PEC wall. The FDTD grid is of 300×240 cells. The FDTD grid spacing is $\lambda/20$ and the time step is set to $dt = 0.99 \cdot dx / (c_0 \sqrt{2.0})$ to satisfy CFL stability criteria [7].

A PEC interface is placed at the center of the waveguide at $x = 150$. An aperture is created by creating a slit by removing the PEC cells [5]. The waves are blocked by the PEC walls, so the waves cannot pass through the PEC wall. However, the waves pass through the slit section [5].

As per Huygens–Fresnel Principle, the waves passing through the slit aperture act as a secondary radiating wave source [6].

CPML is an advanced boundary absorption which helps to absorb the waves so that no reflection takes place. The layers gradually absorb the wave and only minimal waves are reflected back. The CPML is applied at the boundaries and helps to absorb waves at various incident angles.

The CPML is a basic one and follows the paper [1]. The PML is 20 cells thick. The σ in the code refers to conductivity [1]. The waves entering the PML layer experience a gradual increase in conductivity, i.e. the conductivity is very small in the interior and is maximum at the outer edge so that the waves do not get an impedance mismatch due to sudden conductivity change. The higher the conductivity is, the stronger the absorption will be [1].

There are other variables like κ which helps to stretch the space so that waves will have some space to slow down, and α which helps to act as a damper improving the absorption [1].

The ψ arrays are the CPML memory variables (the terms with ψ in the code such as $\psi_{Ez_x_low}$) which aid in storing the earlier field derivative information [1]. Those values are used as correction terms and are subtracted in the field update, removing the energy and creating absorption rather than reflection.

For the measurement of the field and checking the decay, a measurement

point is placed inside the PML. We can see how the magnitude will decay.

Plot 1: 2D near-field E_z snapshot showing cylindrical wave radiation from the aperture

A PEC wall is setup at the center of the cell at $x = 150$. The plot shows the Huygens–Fresnel principle of wave acting as a secondary source at the slit. The interference pattern before the PEC wall is where there is the incident and reflection of the wave. As the wave passes through the slit, the waves diffract and create a cylindrical wavefront.

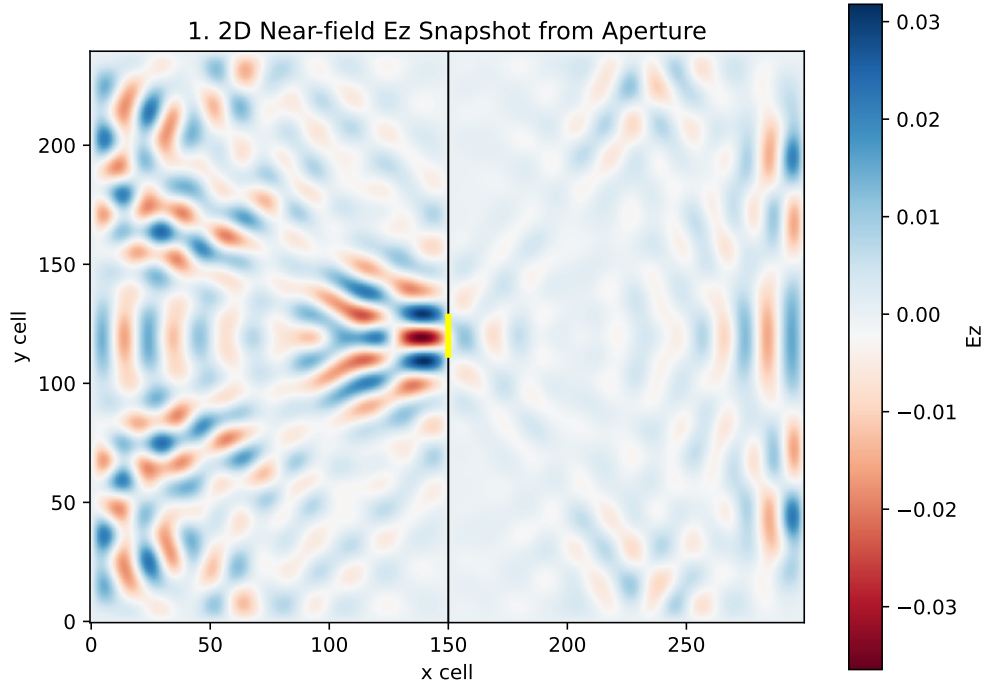


Figure 29: 2D near-field E_z snapshot showing cylindrical wave radiation from the aperture

Plot 2: The E -field amplitude across aperture plane

The PEC wall along the y axis is from $y = 0$ to 112 and from $y = 128$ to 240. From the plot, we can clearly see that the E field magnitude is 0. As we

know, the electric field in the PEC is 0 which tells that the PEC is working. The red dashed line represents the slit opening. The blue curve represents energy passing the wall.

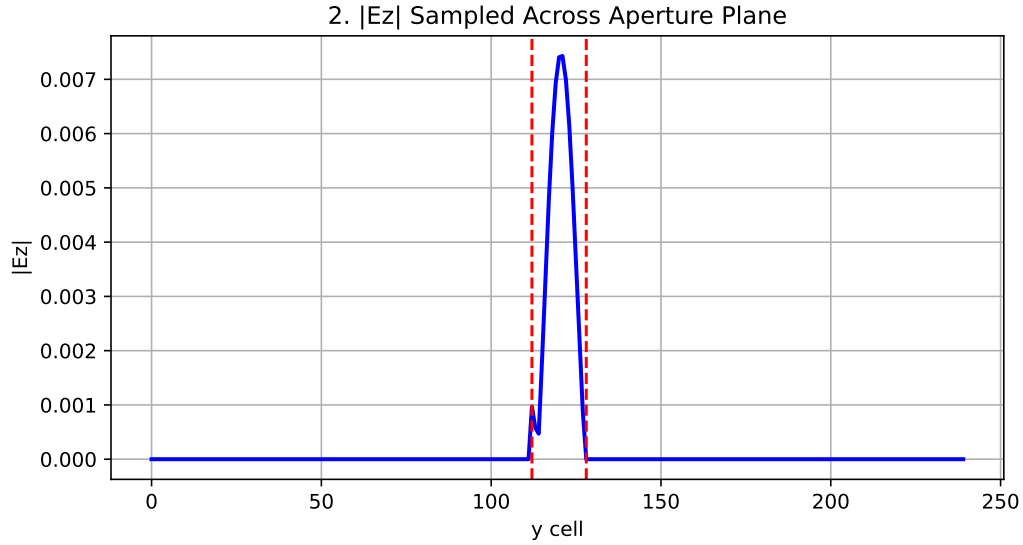


Figure 30: The E -field amplitude across aperture plane

Plot 3: Far-field radiation pattern

This provides the radiation pattern of the field by sampling the E field at a fixed point. The plot indicates that the wave is not uniform in all directions and we can see main lobes and side lobes. The multi-lobe pattern is due to the slit being very small and not allowing the energy to propagate as a narrow beam.

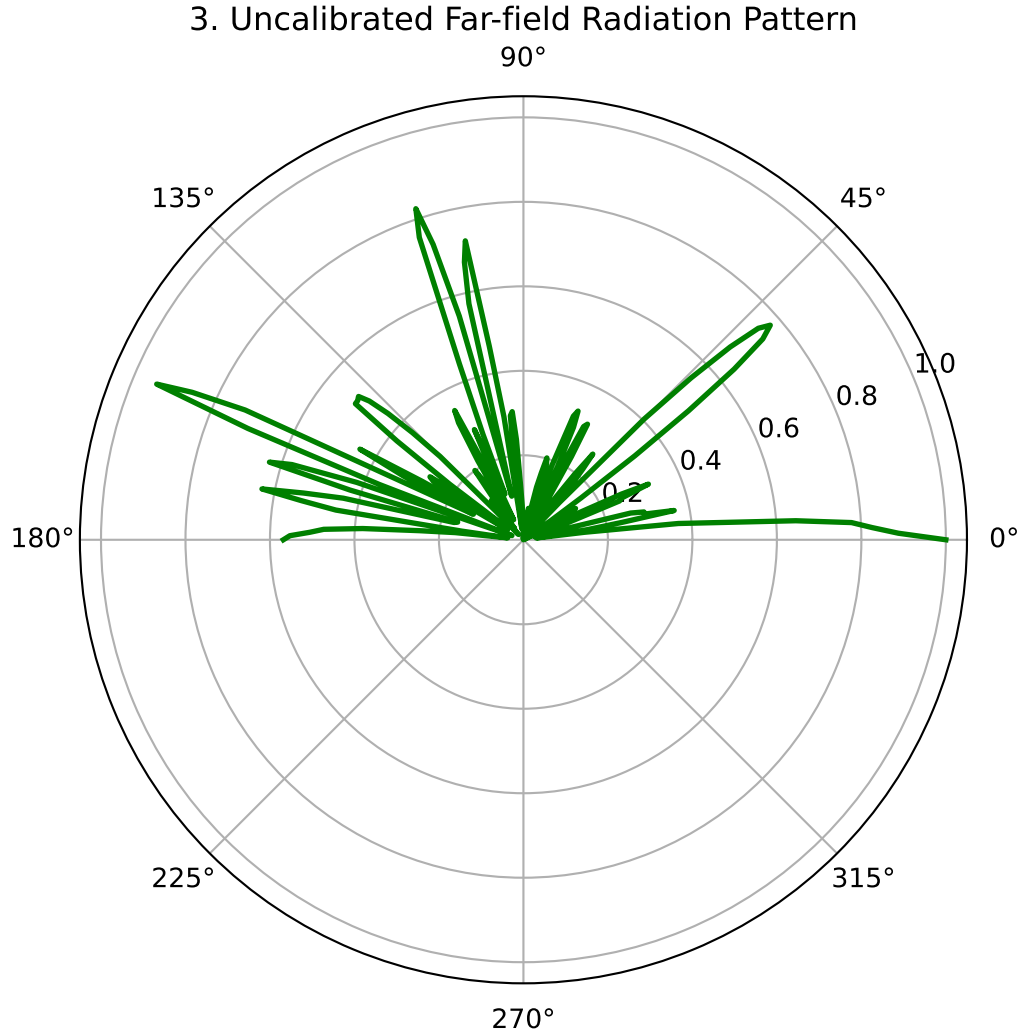


Figure 31: Far-field radiation pattern

Plot 4: E_z field deep inside PML

The simulation uses a probe inside the PML to measure the field value. PML is 20 cells thick and the probe is placed at 18 cells, deep inside the PML. The E_z values are displayed in logarithmic scale. We can see there is 0 magnitude until it hits 0.6 ns. This is because the wave needs some time to reach the probe from the source. The wave drops and goes to around 10^{-5} and 10^{-6} .

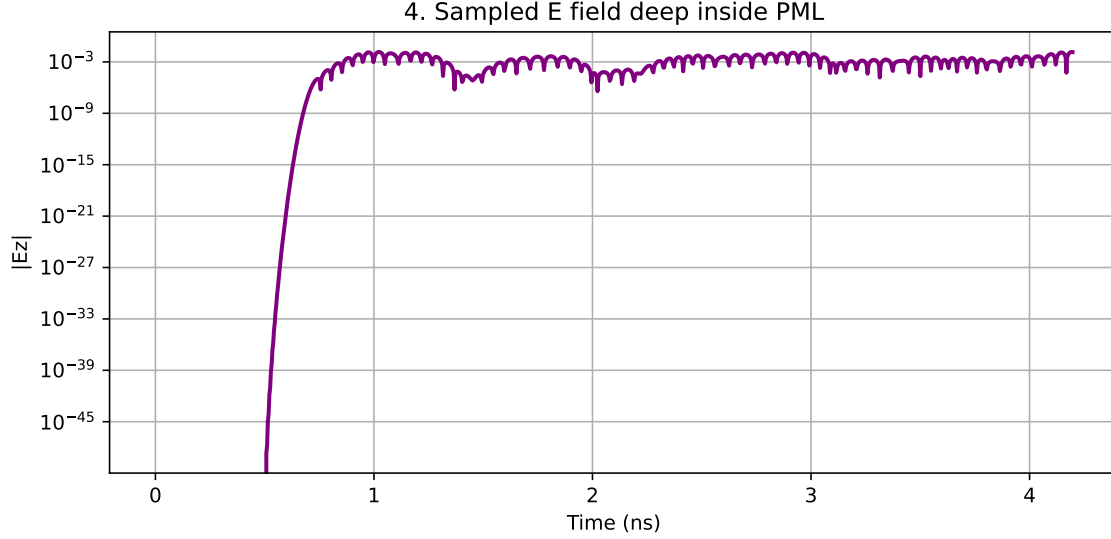


Figure 32: E field deep inside PML

Problem 8: 2D H-Plane Flared Horn Antenna

Description: Combine waveguide structures and radiation boundaries. Simulate a parallel- plate waveguide that gradually flares outward into a horn. Feed the waveguide with the fundamental mode. The surrounding domain must be terminated with a robust PML.

→

Here, the FDTD simulation consists of a flared horn antenna. A waveguide is used which gradually flares in outward direction into a horn shape. Such configuration can be used to see how a wave transitions from a standard waveguide to a flared horn antenna [3].

The same CPML used in Problem 7 has also been used here to absorb the waves and effectively reduce reflections. The same conductivity σ , stretching factor κ , and damping factor are used for the PML [1].

The source is excited with a soft source frequency modulated pulse. Horn antenna shape is structured and PEC is implemented using a boolean mask outside the horn walls, to create the desired horn geometry [4]. The waveguide helps guide the waves into the horn structure.

The first loop is where the shape of the horn antenna takes place. It checks through the waveguide from the waveguide starting point to the aperture. It also checks if the position is before the flare position or after the flare position and creates the shape accordingly.

The waveguide starts at $x = 30$ and the flare, which is the part where the shape starts to go outwards, starts at $x = 120$. The aperture is the end part of the horn antenna which is positioned at $x = 300$. The electric field near the aperture is stored over time and FFT is used to obtain the frequency component at 10 GHz. It is then used to calculate the phase across the aperture [3].

The FDTD grid spacing is $\lambda/20$ and the time step is set to $dt = 0.99 \cdot dx/(c_0\sqrt{2.0})$ to satisfy CFL stability criteria [7].

The PEC mask is used to create the back side PEC and cuts an opening inside that PEC which forms the waveguide.

Plot 1: 2D E -field snapshot showing the wave transitioning from the waveguide into the flare

The plot shows the wave transitioning from the narrow waveguide into the horn flare. Inside the waveguide, the wave stays confined and as the wave enters the horn flare which expand outwards, the wavefront spreads and radiate outward. Also, we do not see any reflected waves or interference due to waves being absorbed.

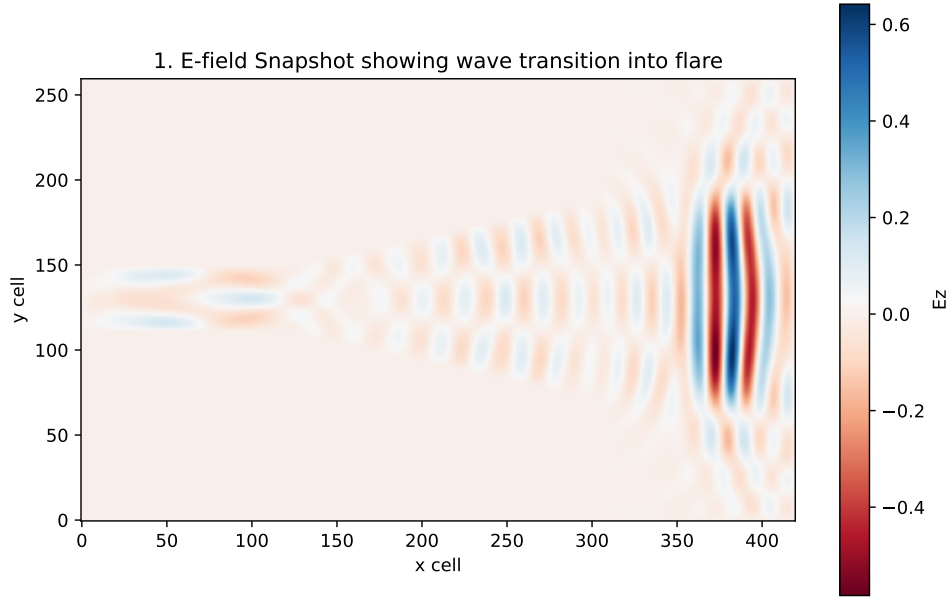


Figure 33: 2D E -field snapshot from the waveguide

Plot 2: 2D E -field snapshot of the wave radiating from the aperture into the free space domain

The snapshot shows the wave leaving the horn aperture into the free space. If we see at the aperture, we observe that the wave is getting a fading pattern and is spreading out as it propagates away from the horn aperture. Similarly, we do not see any reflections here from the boundaries.

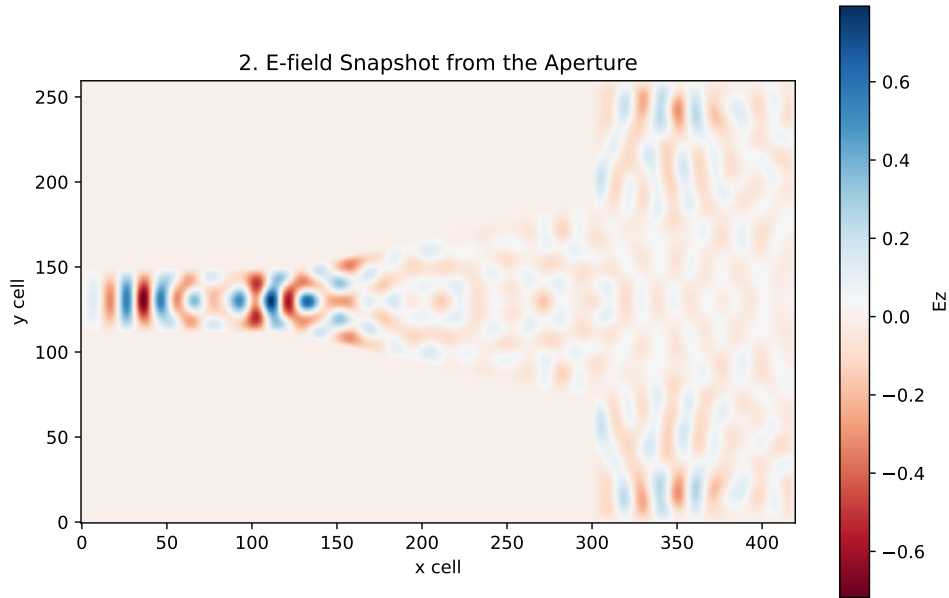


Figure 34: 2D E -field snapshot radiating from the aperture

Plot 3: The normalized radiation pattern (polar plot)

The polar plot provides the antenna directivity with a normalized radiation pattern. The main lobe is narrower and more concentrated in the forward direction compared to the problem 7 slit pattern where the energy is more spread. This means the horn antenna has improved directivity compared to the slit geometry.

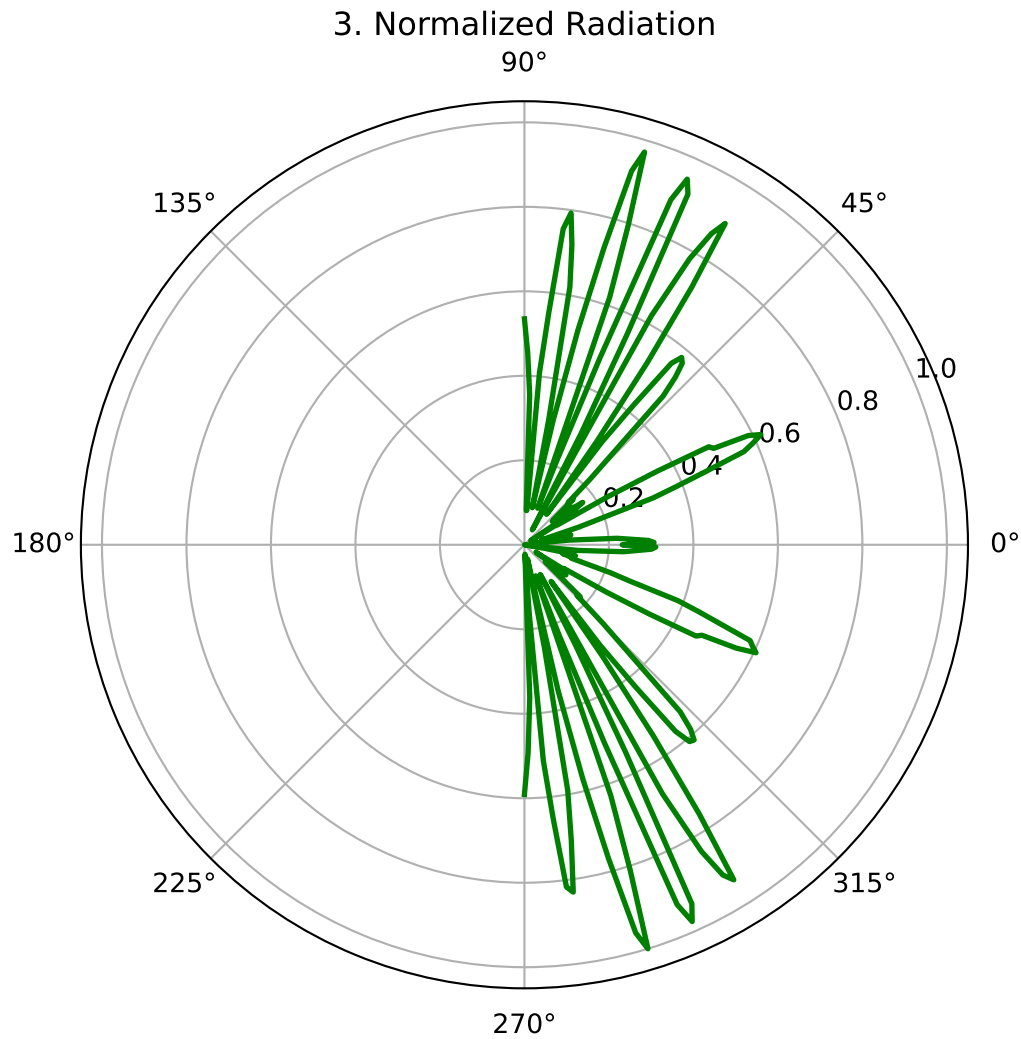


Figure 35: Normalized radiation pattern

Plot 4: Phase distribution of the electric field across the broad aperture of the horn

The plot shows the phase distribution of the E field across the horn aperture. The inverted U shape at the center is the quadratic phase error. When the wave travels from the waveguide to the horn flare, the wave travelling at the edges need to have a longer distance than the waves at the center. This lag is seen as the quadratic phase error.

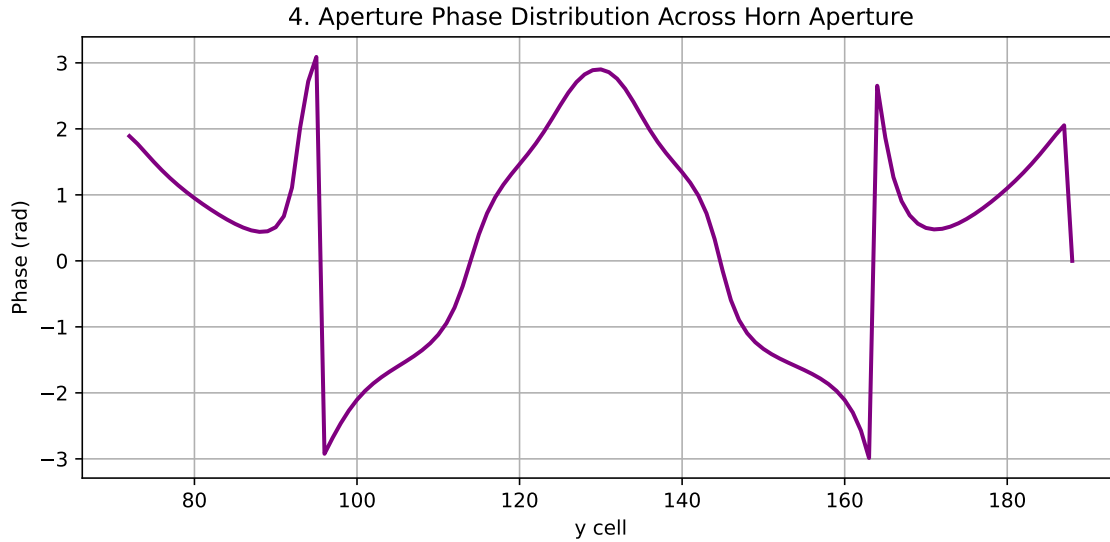


Figure 36: Phase distribution of the electric field

Github Repo Address

https://github.com/bikesh10-npl/ECE_5040_EM_Simulation_Spring_2026_BS

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